

SILK++

THE KILL-LIST BATTLE PLAN

A-to-Z, date-by-date master blueprint to build the world's most emotionally expressive, lowest-latency voice AI.

TARGETS: ELEVENLABS / HUME EVI 3 / RUMIK SILK / SARVAM BULBUL

TARGET SPECS (what we must beat)

METRIC	ELEVENLABS	HUME EVI 3	RUMIK SILK	SARVAM	SILK++ TARGET
First-packet latency	~75 ms	~300 ms	~1000 ms	~100 ms	<= 80 ms
End-to-end voice-to-voice	~500 ms	~1200 ms	<1000 ms	~600 ms	<= 350 ms
UTMOS naturalness	4.05	4.10	~3.95 (est)	~3.85	>= 4.25
Code-switch (Hinglish)	Weak	Poor	Best in mkt	Good	SOTA
Emotion control	Tags	Implicit+RL	Inline tags	Basic	Hierarchical+RLHF
Zero-shot voice clone	5s	Persona-only	Unknown	Limited	3s
Cost / min TTS	\$0.11	Premium	\$0.02-0.03	\$0.03	\$0.008
Languages	32	~10	8 Indic	11 Indic	22+ (10 Indic, 12 Intl)

Mission: Win.

16 weeks. 7 people. One goal: own the emotional voice AI category.

Executive Briefing

The thesis. Voice AI in 2026 is a four-horse race: ElevenLabs owns latency, Hume owns emotional depth, Rumik Silk owns Indic code-switch, Sarvam owns the Indian enterprise pipe. None of them own **all four**. The opportunity is to build a single model that beats each incumbent at its own strongest axis — and ships in 16 weeks.

The weapon. SILK++ is a dual-transformer, autoregressive speech-language model trained on a custom Mimi-Lite RVQ codec, conditioned via hierarchical emotion control vectors, post-trained with DPO + GRPO on 100K human-rated prosody pairs, and served through an 80ms chunked streaming kernel. Target: 80ms first-packet latency, 4.25 UTMOS, native Hinglish / Tanglish / Manglish / Kanglish, 3-second zero-shot voice clone.

The why-now. Sesame open-sourced CSM-1B. Kyutai open-sourced Mimi. Fish-Audio open-sourced S2-Pro with GRPO recipes. EmoVoice proved natural-language emotion control works. The full ingredient list for a Hume-killer is already on the shelf. Whoever **assembles fastest** wins.

TIMELINE

16 wk

ship D-day 03 Sep 26

HEADCOUNT

7

4 ML + 1 data + 1 infra + 1 audio

BUDGET

\$520K

compute + actors + tooling

LATENCY

80 ms

first-packet, beats ElevenLabs Flash

The Three Knives We Drive Into the Incumbents

- 1. Phonetic Code-Switch Tokenizer (PCS-Tok)** — unify Latin-script and native-script Indic phonemes so the model treats 'main kal milunga' and '■■■■ ■■ ■■■■■■■■■■' as the same distribution. Silk only solved this partially. We solve it at the tokenizer level.
- 2. Hierarchical Emotion Control (HEC)** — discrete tags + continuous intensity + duration vectors injected via FiLM modulation on every backbone layer. Hume EVI 3 nails emotion understanding; we match it on understanding and beat it on *controllable expression*.
- 3. DPO + GRPO Prosody Alignment** — 100K human preference pairs, two-stage preference optimization. Silk shipped on SFT alone. This is the asymmetric quality unlock.

Know your targets.

What each incumbent is, where they bleed, and how we exploit it.

Enemy Dossiers

ElevenLabs / The Latency King

What they are. Flash v2.5 hits ~75ms model inference latency, 32 languages, the de-facto TTS API standard. v3 model is higher quality but slower. They serve from 3 global clusters.

Where they bleed. (a) **Price** — ~\$0.11/min puts them out of reach for Indian consumer scale. (b) **Hinglish is weak** — they treat Romanized Hindi as English with funny phonemes. (c) **Emotion is tag-only**, no nuanced control. (d) PVC voices add latency.

How we cut. Match 75ms latency with our Mimi-Lite + 8-codebook fast mode, beat them on Hinglish by ~30 UTMOS points, ship at \$0.008/min via self-hosted H100 inference.

Hume EVI 3 / The Empathy Layer

What they are. Single autoregressive model handling transcription + language + speech (voice-to-voice tokens). Trained on millions of human vocal modulations. Beats GPT-4o blind on empathy, expressiveness, emotion understanding. RL-trained to hone preferred voice qualities. Practical latency ~1.2s end-to-end.

Where they bleed. (a) **English-centric**, mediocre in Indic. (b) **Closed model**, no on-prem. (c) **\$\$\$** and US-only deployment. (d) Practical latency still >1s — they admit it.

How we cut. Build the same speech-language token architecture (we have the Sesame CSM blueprint), train it on 80K hours of Indic + English, drive practical latency to 350ms via co-located inference in Mumbai + Singapore, expose hierarchical emotion control to developers.

Rumik Silk / The Hinglish Specialist (you)

What it is. Fine-tuned dual-transformer expressive TTS over Mimi-like RVQ. Code-switched Hinglish/Tanglish/Manglish/Kanglish, inline emotion tags (<laugh>, <warm tone with light jolt>), sub-1s response. Cost ~\$0.02-0.03/min. Built from a fork of CSM/Fish-Speech base, hard data engineering on Romanized Indic corpus.

Where it bleeds (per its own research blog). (a) Multi-speaker generalization was hard, implies dataset speaker diversity is still limited. (b) Long-form coherence ~2 min. (c) No DPO/GRPO post-training. (d) No zero-shot voice clone in public API yet.

How we cut. SILK++ is Silk + DPO + 5x speaker diversity + PCS-Tok + 5-sec zero-shot clone + edge-deployable 8-codebook mode. Same Indic strength, 2x better naturalness, half the latency.

Sarvam Bulbul v3 / The Enterprise Indic Play

What they are. 22 Indian languages, 25+ voices, enterprise contracts with GoI, Microsoft Azure distribution partner. Real-time streaming, emotion control claimed.

Where they bleed. (a) Voice quality is functional, not delightful — UTMOS ~3.85 vs ElevenLabs 4.05. (b) **No code-switching** — they treat each language as separate. (c) Strong B2B distribution but weak consumer voice. (d) No emotional depth.

How we cut. Don't compete on enterprise B2B (their moat). Compete on consumer companion / EdTech / entertainment where natural emotion and Hinglish code-switch matter most.

The full stack.

Codec, backbone, decoder, speaker conditioning, streaming kernel.

The SILK++ Architecture

Three transformers, one codec, one streaming kernel. The architecture is a fusion of Sesame CSM (dual-transformer backbone + audio decoder), Kyutai Mimi (split-RVQ codec), VoXstream (incremental phoneme transformer for sub-100ms first-packet), and EmoVoice (LLM-native emotion control).

LAYER	COMPONENT	PARAMS	ROLE	SOURCE
L0	Mimi-Lite Codec	180M	24kHz audio → 12.5Hz RVQ tokens, 32 codebooks	Kyutai Mimi
L1	PCS-Tok BPE	—	Unified Latin + Devanagari + 9 Indic scripts, phonetic anchors	Custom
L2	Phoneme Transformer	120M	Incremental, dynamic look-ahead — VoXstream-style	VoXstream 2509.15969
L3	Slow Backbone (Temporal)	8B	Predicts cb0 (semantic) at 12.5Hz, conditions everything	Sesame CSM + Llama-3
L4	Fast Decoder (Depth)	350M	Predicts cb1..cb31 in parallel per frame	Sesame CSM
L5	Emotion FiLM Heads	30M	Inject discrete tag + continuous intensity + duration	EmoVoice 2504.12867
L6	Speaker Encoder	20M	ECAPA-TDNN + WavLM fusion → 256-dim cond vector	ReDimNet / WavLM
L7	Streaming Kernel	—	vLLM custom audio sampling, 80ms packets, KV reuse	Custom + vLLM

Token Flow (one inference step)

```
[text input] → PCS-Tok BPE → text tokens (T)
[5s speaker ref] → ECAPA + WavLM → spk_emb (256-d)
[T + spk_emb + emotion_ctrl] → Slow Backbone (8B) → hidden state h_t, cb0_t
[h_t + cb0_t + spk_emb] → Fast Decoder (350M) → cb1..cb31 in 1 forward pass
[cb0..cb31] → Mimi-Lite Decoder → 80ms audio @ 24kHz
Total per-frame compute: ~12 ms on H100 → first packet ≤ 80 ms
```

Why dual-transformer wins

- bullet Slow backbone (8B) only runs on cb0 → semantic / prosody live in cb0 → most of the compute pays for naturalness, not bitstream fidelity.
- bullet Fast decoder (350M) runs 1/16 of the time via Sesame's compute amortization → 16× memory savings during training.
- bullet Splitting semantic vs acoustic codebooks lets us drop 24 acoustic codebooks at inference for edge mode (8-codebook → mobile / wearables).
- bullet Mimi-Lite at 12.5 Hz gives 80ms-aligned frames natively → streaming protocol falls out for free.

Why we add an incremental Phoneme Transformer

Pure CSM backbones wait until they have a sentence boundary before flushing audio. VoXstream proved you can hit 102ms first-packet by feeding phonemes incrementally with dynamic look-ahead. We fold this in front of our slow backbone so the very first word streams while the rest of the sentence is still being tokenized.

Data is the moat.

60% of project time. Where Silk wins and where we will out-curate them.

Data Strategy — The Real Moat

Truth bomb. The architecture for SILK++ is mostly open-source remixing. The moat is data — specifically the code-switched, emotion-annotated, multi-speaker Indic corpus. Budget 60% of project hours here. This is where you out-execute Silk and Hume.

TIER	USE	HOURS	SOURCES	STATUS
T0 Codec pretrain	Mimi-Lite	50,000	LibriHeavy, Emilia-EN, Multilingual LibriSpeech, AudioSet (Indian subset)	Public
T1 Backbone pretrain	Slow + Phoneme TF	80,000	T0 + Shrutilipi, Vaani, IndicVoices, IndicTTS, Common Voice IN, AI4Bharat	Public
T2 Code-switch SFT	The Silk killer	3,000	Synthetic: GPT-4o-mini generates Hinglish/Tanglish dialogues → Fish-S2 + ElevenLabs synthesis	Build
T3 Emotion SFT	HEC training	1,500	EmoVoice pipeline + 50 paid Indian voice actors × 20h studio recording across 9 emotion classes	Build
T4 DPO pairs	Prosody alignment	100K pairs	Scale AI / Surge / Karya Inc — 5 raters × pair, fleiss-kappa > 0.6 retention	Buy
T5 Eval set	Daily regression	20 hours	Held-out from T1-T3, plus 200 Hinglish news, 200 EdTech transcripts, 200 companion-app dialogues	Build

The Code-Switch Data Engine (your real superpower)

This is the asymmetric advantage. Build a 3-stage synthetic pipeline that produces ~3,000 hours of code-mixed audio in 6 weeks at <\$30K compute cost.

```
# Stage A — Dialogue generation
GPT-4o-mini prompt: 'Generate 500 lines of natural Mumbai gen-Z
Hinglish dialogue mixing Latin-script Hindi with English code-switches,
with [emotion=excited|sad|sarcastic] tags interleaved naturally.'

# Stage B — Voice synthesis
Synthesize each line in 3 different voices using
Fish-Speech S2-Pro (free) and ElevenLabs Multilingual v2 (premium).
Yield: 3,000 hrs raw × 3 voices = 9,000 hrs candidate.

# Stage C — Quality filter
1. Whisper-large-v3 ASR back-translation → WER < 8%
2. UTMOS predictor → score > 3.8
3. DNSMOS-P808 SNR > 25 dB
4. Human spot-check 1% sample → reject batch if <90% accept
Survival rate: ~33% → 3,000 hrs clean code-switch corpus.
```

The Voice Actor Studio Plan

- bullet** Hire 50 actors via Backstage India + IndianTalent agencies — diverse: 25M / 25F, ages 18-55, 8 regional accents (Mumbai, Delhi, Kolkata, Chennai, Bengaluru, Hyderabad, Punjab, Kerala).
- bullet** Each actor records 20 hours over 4 weeks at a Mumbai/Bengaluru studio (NeoSound, Sound Ideaz) — split as 8h neutral + 12h emotional (9 emotion classes × ~1.3h each).

bullet Scripts blend: 40% Hinglish dialogue (curated from social media + film), 30% news/educational, 20% companion-app conversational, 10% edge cases (whisper, yelling, laughter).

bullet Per-actor cost: ~\$1,500 fee + studio time. Total: ~\$95K. Yields the highest-margin 1,500 hours of training data on the planet.

GPUs, storage, orchestration.

What to buy, where to host, how to scale.

Hardware, Infra, MLOps

Training cluster

RESOURCE	SPEC	QTY	PROVIDER	MONTHLY COST
Training GPUs (Stage 2-7)	H100 SXM 80GB	64	CoreWeave reserved (commit 3mo)	\$130K
Codec training (Stage 1)	H100 SXM 80GB	8	Lambda Labs on-demand	\$22K (14d burst)
Data pipeline	CPU c7i.24xl + S3	8 + 50TB	AWS Mumbai (ap-south-1)	\$9K
Inference (prod)	H100 PCIe + L40S mix	16+24	CoreWeave + Vultr Mumbai	\$48K
Edge / mobile testbed	Snapdragon 8 Gen 3 dev kits, Apple M3Max	3	Direct purchase	\$25K one-time
Dev workstations	RTX 6000 Ada	7	Direct purchase	\$70K one-time

MLOps stack — no improvisation

- Experiment tracking:** Weights & Biases Teams plan (every run logged with audio samples).
- Code repo:** Monorepo on GitHub with Pants build, branch protection, signed commits.
- Data:** Lhotse manifests + WebDataset shards on S3, versioned with DVC.
- Training:** PyTorch 2.5 + FSDP-2 + FlashAttention-3, torch.compile mandatory.
- Serving:** vLLM 0.6+ with custom audio sampling kernel; Triton frontend for batching.
- Observability:** Prometheus + Grafana + custom 'latency budget tracer' dashboard.
- Eval:** Nightly regression run on T5 eval set; alarms on UTMOS / WER / latency regression.

Inference geography (the latency unlock)

ElevenLabs gives 75ms *model* latency but +60ms network from India users. Hume serves from US West Coast — adds 200ms RTT to Mumbai. Our edge: **co-locate inference in Mumbai (ap-south-1) and Singapore (ap-southeast-1)**. RTT to 90% of South-Asia traffic is <25ms. Combined with 60ms model latency we hit **practical 85-100ms TTFA** — best in class for Indian users.

Date by date. Owner by owner.

Issued 14 May 2026. D-day 03 Sep 2026.

The A-to-Z 16-Week Plan

Read this like a war plan. Each week has a theme, deliverables, owners (codenames), and a 'kill metric' — the number that must move that week or we re-plan. Calendar dates are anchored to the issue date of this document (14 May 2026). All times IST.

OWNER	ROLE	RESPONSIBILITY
VB	CTO / You	Strategy, hiring, fundraise comms, final architecture sign-off
ML-LEAD	Principal ML Engineer	Backbone + decoder training, curriculum management
CODEC	Audio Researcher	Mimi-Lite codec, vocoder, audio quality eval
DATA	Data Engineer	Pipelines, manifests, synthetic data engine, actor studio coord
INFRA	Infra / MLOps Engineer	Cluster, K8s, vLLM serving, observability
RLHF	Alignment Researcher	DPO/GRPO, preference data ops, reward model
FE	Streaming Engineer	WebSocket server, mobile SDK, latency optimization

PHASE

PHASE 1 / FOUNDATION

WEEK 1 | 14–20 May 2026 | **Hire, set up cluster, fork CSM** | Lead: **VB + INFRA**

Kill metric: Cluster online with 8 GPUs benchmark-running CSM-1B inference by Friday EOD.

- bullet** VB: Close ML-LEAD offer (target: ex-DeepMind / Sesame / Fish-Audio); send INFRA + DATA offers.
- bullet** VB: Sign CoreWeave 3-month H100 reservation (64 GPUs); sign AWS Mumbai ap-south-1 commit.
- bullet** INFRA: Stand up cluster, Slurm + K8s hybrid, S3 bucket structure, W&B; Teams workspace.
- bullet** INFRA: Fork sesame/csm-1b, fishaudio/fish-speech, kyutai-labs/moshi (for Mimi reference).
- bullet** VB: Outline founders-only Notion doc — kill-metric dashboard auto-syncing from W&B.;

WEEK 2 | 21–27 May 2026 | **Data team online + first synthetic batch** | Lead: **DATA**

Kill metric: First 100K synthetic code-switch lines generated and audio-synthesized by Friday.

- bullet** DATA: Pull Tier 0 corpora (LibriHeavy, Emilia, MLS) to S3 ap-south-1 — ~6TB.
- bullet** DATA: Pull Tier 1 Indic (Shrutilipi, Vaani, IndicVoices, AI4Bharat) — ~4TB.
- bullet** DATA: Stand up Lhotse manifest pipeline; first 10K-hour clean manifest ready.
- bullet** DATA + VB: Draft GPT-4o code-switch dialogue prompt set (1,000 templates × 5 emotion modes).
- bullet** ML-LEAD onboard: Day 1, ship a benchmark report on CSM-1B Hinglish baseline (WER + UTMOS).

WEEK 3 | 28 May – 03 Jun 2026 | **PCS-Tok tokenizer build** | Lead: **ML-LEAD + DATA**

Kill metric: PCS-Tok BPE achieves <8% reconstruction error on held-out Hinglish text; vocab size 64K.

- bullet ML-LEAD: Train custom BPE tokenizer on 10M lines unified (Latin + Devanagari + Tamil + Telugu + Kannada + Malayalam + Bengali + Gujarati + Punjabi + Marathi).
 - bullet ML-LEAD: Build epitran + IndicXlit phonetic mapping table; inject as auxiliary phoneme stream.
 - bullet DATA: Synthetic engine Stage A producing 50K lines/day; Whisper QA running in parallel.
 - bullet CODEC onboard: ramp on Kyutai Mimi codebase; reproduce 24kHz reconstruction PESQ ≥ 3.9 on LibriTTS.
 - bullet VB: Hire RLHF lead (target: ex-Anthropic / OpenAI alignment); 4 of 7 seats filled.
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WEEK 4 | 04–10 Jun 2026 | **Mimi-Lite codec training starts** | Lead: **CODEC**

Kill metric: Mimi-Lite passes Day-3 sanity: PESQ ≥ 3.5 on dev set; commit loss converging.

- bullet CODEC: Mimi-Lite (180M, 32 codebooks, 12.5Hz, 24kHz, split-RVQ) starts pretraining on 8×H100, 14-day budget.
 - bullet CODEC: Implement RVQ-dropout: at each batch, sample $k \in [8, 32]$ for codebook truncation → enables 8-codebook fast mode at inference.
 - bullet ML-LEAD: Modify CSM-1B vocab layer to accept PCS-Tok tokens + emotion special tokens (256 reserved IDs).
 - bullet DATA: Code-switch corpus crosses 500K filtered lines; WER target <8% holding.
 - bullet FE onboard; ramps on vLLM internals + Mimi streaming decoder.
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PHASE

PHASE 2 / PRETRAIN & FINE-TUNE

WEEK 5 | 11–17 Jun 2026 | **Backbone fine-tune kickoff (English first)** | Lead: **ML-LEAD**

Kill metric: Stage-2 English-only backbone hits UTMOS 4.0 on held-out LibriTTS by week-end.

- bullet** ML-LEAD: Init backbone from sesame/csm-3b weights; replace tokenizer + audio token embeddings.
- bullet** ML-LEAD: Stage-2 continued pretrain — English-only Tier-1, 50K steps, LR 1e-5, 64×H100.
- bullet** CODEC: Mimi-Lite codec passes acceptance (PESQ $\geq$ 4.0, STOI $\geq$ 0.95); freeze checkpoint.
- bullet** DATA: Tier-3 emotion corpus design freeze; actor studio bookings (NeoSound Mumbai) confirmed for Week 7.
- bullet** RLHF onboard; begins reading SpeechAlign + Fish-S2 GRPO papers; writes preference-data SOP.

WEEK 6 | 18–24 Jun 2026 | **Multilingual continued pretrain** | Lead: **ML-LEAD**

Kill metric: Multilingual backbone UTMOS 4.05+ on English, $\geq$ 3.7 on Hinglish/Tamil dev sets.

- bullet** ML-LEAD: Stage-3 multilingual continued pretrain — full Tier-1, 200K steps, LR 1e-5 $\rightarrow$ 5e-6 linear decay (the 'gradual adaptation' lesson from Silk's blog).
- bullet** INFRA: Build streaming inference prototype on top of vLLM; first 80ms-chunked WebSocket demo, measured TTFA 180ms (target 80ms).
- bullet** DATA: Tier-2 code-switch corpus crosses 1,500 hours filtered; 2,000-line human spot-check passes 92% accept.
- bullet** FE: Mobile-side prototype — ONNX export of Mimi-Lite 8-codebook fast mode; Pixel 8 inference @ 1.3× real-time.
- bullet** VB + RLHF: Sign Surge/Scale contract for 100K DPO preference pair collection (delivery Wk 13).

WEEK 7 | 25 Jun – 01 Jul 2026 | **Code-switch SFT + actor recording starts** | Lead: **ML-LEAD + DATA**

Kill metric: Code-switch SFT model speaks Hinglish without script-confusion artifacts; A/B vs Silk demo shows $\geq$ parity on internal 50-line eval.

- bullet** DATA: 50 voice actors begin studio recording (NeoSound Mumbai + Sound Ideaz Bengaluru); 12.5h/actor/wk pace.
- bullet** ML-LEAD: Stage-4 Code-Switch SFT — 80K steps, LR 5e-6, full Tier-2, with speaker-contrastive aux loss.
- bullet** RLHF: Reward model design freeze — WavLM-large features $\rightarrow$ 3-layer MLP $\rightarrow$ scalar prosody score; uses 5K seed pairs internally rated.
- bullet** FE: Phoneme Transformer (VoXstream-style) prototype integrated — incremental decoding hits 110ms TTFA on dev.
- bullet** INFRA: Multi-region inference plan locked — Mumbai + Singapore + Frankfurt (for Europe expansion later).

WEEK 8 | 02–08 Jul 2026 | **Emotion SFT + first alpha demo** | Lead: **ML-LEAD + CODEC**

Kill metric: Closed-alpha demo demonstrates clear emotion control on 5 demo dialogues. UTMOS $\geq$ 4.10. WER $\leq$ 9% Hinglish.

- bullet** ML-LEAD: Stage-5 Emotion SFT — 40K steps with HEC (hierarchical emotion control) — discrete tag tokens + continuous intensity FiLM injection.
- bullet** CODEC: Vocoder polish pass — HiFi-GAN-v3 head re-trained on emotion-labelled corpus for better laughter / whisper rendering.
- bullet** DATA: Actor corpus crosses 700 hours; 9 emotion classes balanced; transcripts + prosody alignments uploaded.

- bullet** INFRA: First end-to-end alpha demo (closed) — VB shows internal team a Hinglish companion app demo with 4 emotions.
 - bullet** VB: Investor update sent (Series-A signal-light); booked 6 strategic intros for Wk 14.
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PHASE

PHASE 3 / ALIGNMENT & STREAMING

WEEK 9 | 09–15 Jul 2026 | **Preference data collection live** | Lead: **RLHF + DATA**

Kill metric: Reward model AUC ≥ 0.75 . Streaming TTFA $\leq 100\text{ms}$. 20K preference pairs collected, on pace.

- bullet** RLHF: 100K (prompt, audio_A, audio_B) preference pairs collection begins — 5 raters/pair, Karya Inc. + Surge AI, fleiss-kappa target ≥ 0.6 .
- bullet** RLHF: Reward model v1 trained on 5K seed pairs; AUC ≥ 0.75 against held-out human votes.
- bullet** ML-LEAD: Long-context experiment — extend context to 5 min via NTK-aware RoPE scaling; perplexity holds.
- bullet** FE: Streaming kernel hits 95ms TTFA on H100; KV-cache reuse across turns cuts continuation to 45ms.
- bullet** INFRA: Cost dashboard live — current run rate at \$480K projected vs \$520K budget; on plan.

WEEK 10 | 16–22 Jul 2026 | **DPO alignment** | Lead: **RLHF + ML-LEAD**

Kill metric: DPO post-training improves win-rate vs Stage-5 by $\geq 65\%$ on held-out human-judged eval (n=200).

- bullet** RLHF + ML-LEAD: Stage-6 DPO run — 10K steps, $\beta=0.1$, LR $1e-7$, $16\times H100$, 1 day budget. Two ablations vs reference model.
- bullet** DATA: Synthetic engine retired; T2 corpus closed at 3,200 hrs final filtered.
- bullet** CODEC: Edge mode validated — 8-codebook Mimi-Lite reconstructs at PESQ 3.75 (acceptable); on-device Pixel 8 inference $0.85\times$ RT.
- bullet** VB + FE: Closed-beta plan — 100 hand-picked users (Indian creators, Reddit, X) recruited; consent + NDAs filed.
- bullet** RLHF: GRPO recipe locked for Wk 11; reward shaping with secondary intelligibility penalty.

WEEK 11 | 23–29 Jul 2026 | **GRPO polish + closed beta launch** | Lead: **RLHF + FE**

Kill metric: Closed beta p50 TTFA $\leq 90\text{ms}$; thumbs-up $\geq 70\%$; ≥ 30 users return for ≥ 3 sessions in first week.

- bullet** RLHF + ML-LEAD: Stage-7 GRPO — 5K steps, with reward model + intelligibility aux + secondary diversity penalty.
- bullet** FE: Closed beta deploys to Mumbai region; 100 users live with real-time companion app + Hinglish + 9 emotions.
- bullet** FE: Telemetry on — TTFA p50 / p95 / p99, UTMOS proxy, user thumbs up/down, session length.
- bullet** DATA: 50K preference pairs now in; Karya QA reports kappa 0.64 (above threshold).
- bullet** VB: Begin demo recordings — 12 polished 60-second demo clips for Series-A pitch + launch tweet thread.

WEEK 12 | 30 Jul – 05 Aug 2026 | **Hardening, red-teaming, eval blitz** | Lead: **ALL**

Kill metric: Win-rate vs Silk $\geq 55\%$ on Hinglish, vs Hume $\geq 50\%$ on emotion control, vs ElevenLabs $\geq 50\%$ on overall preference.

- bullet** RLHF: Safety red-team — 500 prompts $\times$ 3 attack types (prompt injection, voice clone abuse, harmful content); patch failures.
- bullet** ML-LEAD: Final eval blitz vs ElevenLabs Flash, Hume EVI 3, Rumik Silk, Sarvam Bulbul on 300-prompt SILK++ eval set.
- bullet** CODEC: Ship 'silkpp-edge' model card — 8-codebook + distilled 1.5B backbone for mobile, $4\times$ real-time on M2.
- bullet** FE: Iterate on closed-beta feedback; ship v0.9 with reduced robotic prosody on long sentences (length >300 chars).

bullet VB + INFRA: Pricing model locked — \$0.008/min self-serve, \$0.005/min enterprise commit.

PHASE

PHASE 4 / SHIP & TAKE THE CATEGORY

WEEK 13 | 06–12 Aug 2026 | **Public preview API** | Lead: **FE + VB**

Kill metric: Preview waitlist $\geq 5,000$ signups by Friday. TTFA p95 $\leq 110\text{ms}$ across both regions.

- bullet FE: Public preview API live (rate-limited 50 RPM/user) — `api.rumik.ai/v2/silkpp/tts` and `voice`.
- bullet FE: SDKs shipped — Python, Node, Swift, Kotlin; OpenAPI spec, Postman collection, 8 starter recipes.
- bullet FE: Docs site live — code samples, latency guide, voice-clone tutorial, safety guidelines.
- bullet VB: Wait-list opens; target 5K signups in first 72h (Twitter/X launch + ProductHunt + HN).
- bullet INFRA: Mumbai region scales to 32 H100; Singapore region added (8 H100 burst-capable).

WEEK 14 | 13–19 Aug 2026 | **Series-A signaling + press** | Lead: **VB**

Kill metric: ≥ 2 term sheets verbal interest. Blog post + benchmark trends on HN front page or 5K X impressions.

- bullet VB: 6 investor meetings booked (Lightspeed, a16z, Peak XV, Accel India, General Catalyst, Iconiq); pitch deck w/ live demo.
- bullet VB: 2 media exclusives — TechCrunch (Indian voice AI startup beats ElevenLabs on price + Hume on emotion) + The Ken (deep dive).
- bullet FE: Real-time API SLA published — 99.5% uptime, $\leq 120\text{ms}$ p99 TTFA, automatic failover Mumbai $\leftrightarrow$ Singapore.
- bullet ML-LEAD: First open research blogpost — 'How SILK++ beats Silk and Hume EVI 3 on the SILK++ benchmark' with reproducible eval kit.
- bullet RLHF: Open eval set (250 prompts) released on HuggingFace — pressure competitors to publish results.

WEEK 15 | 20–26 Aug 2026 | **Enterprise pilots + voice clone v2** | Lead: **VB + FE**

Kill metric: Voice clone wins $\geq 60\%$ blind preference vs ElevenLabs IVC in pilot user test ($n=50$).

- bullet VB: 5 enterprise pilot contracts signed — 2 EdTech (Vedantu/Physics Wallah tier), 2 entertainment (Pratilipi, JioHotstar), 1 gov-tech.
- bullet FE: Zero-shot voice clone v2 ships — 3-sec reference audio (vs 5-sec Silk) — uses ECAPA + WavLM fusion + speaker disentanglement aux loss.
- bullet ML-LEAD: Distillation v1 of 1.5B 'silkpp-fast' student model — 2.5 $\times$ speedup, UTMOS holds within 0.1 of teacher.
- bullet INFRA: Cost per minute drops to \$0.006 self-hosted at current GPU util (60% headroom remaining).
- bullet RLHF: DPO preference pipeline now continuous — 5K new pairs/wk going forward (no more batch collection).

WEEK 16 | 27 Aug – 03 Sep 2026 | **PUBLIC LAUNCH — D-DAY** | Lead: **ALL**

Kill metric: Launch day: $\geq 25\text{K}$ sign-ups, $\geq 500\text{K}$ API calls, $\leq 0.2\%$ 5xx error rate, TTFA p95 $\leq 100\text{ms}$.

- bullet VB: General Availability announcement — 03 Sep 2026, 10am IST, simultaneous on X + LinkedIn + blog + press.
- bullet FE: Production launch — 64 H100 capacity Mumbai + 16 Singapore + 8 Frankfurt; auto-scaling K8s + spot fallback.
- bullet ML-LEAD + RLHF: 'SILK++ Eval Leaderboard' goes live with all 5 competitor scores — open methodology, reproducible.
- bullet VB: D-day demo livestream — 30 min show & tell, 4 use cases (companion, EdTech, entertainment, IVR), Q&A.;

bullet ALL: 24h war room — track sign-ups, latency, on-call rotation, hot-fix capacity reserved.

Code-level specs.

Codec, FiLM emotion, DPO loss, streaming kernel, distillation.

Technical Deep Dives — Code-Level Specs

Deep Dive A — Mimi-Lite Codec Spec

Goal: 24kHz mono → 12.5Hz × 32-codebook RVQ stream → reconstruct at PESQ ≥ 4.0. Sized at ~180M params (encoder 90M + quantizer 10M + decoder 80M). Key tweak vs vanilla Mimi: RVQ-dropout during training enables 8-codebook fast-mode at inference for mobile deployment with only ~0.25 PESQ degradation.

```
# Mimi-Lite - encoder
input: x[B, T] @ 24kHz
x = Conv1D(in=1, out=64, k=7, s=1)(x) # B,64,T
for stride in [4, 5, 6, 8]: # downsample to 12.5 Hz
x = ResidualBlock(stride=stride)(x)
x = TransformerBlock(d=512, layers=8)(x) # context
# Split-RVQ quantizer
z_sem = WavLMSemanticHead(x) # cb0 distilled from WavLM-Large L7
z_acou = RVQ(levels=31, codebook=2048)(x - z_sem) # cb1..cb31
# Training losses
loss = MSTFT(reconstruction) + GAN_disc + commit + sem_distill(z_sem, wavlm_target)
# Trick: RVQ-dropout
k ~ Uniform(8, 32); use only first k codebooks for decode → 8-codebook mode 'for free'
```

Deep Dive B — Hierarchical Emotion Control (HEC)

Three concurrent signals fused into the backbone via FiLM modulation: (1) discrete tag tokens in the prompt — <laugh>, <whisper>, <excited> etc.; (2) continuous intensity scalars in [0,1]; (3) duration hint in milliseconds. The model learns to bind these to the downstream codebook-0 token distribution.

```
# In prompt template
<sys speaker=mum_f02 baseline=cheerful energy=0.7>
<text> arey <laugh intensity=0.6 dur=400ms> tu seriously aaya?!
<whisper intensity=0.8> mujhe yakeen nahi ho raha.

# FiLM injection (every Llama layer)
gamma, beta = MLP_emo([tag_emb, intensity, dur]) # 2 × d_model
h = LayerNorm(h) * (1 + gamma) + beta # FiLM
h = h + SelfAttn(h)
h = h + FFN(h)
# Training: random-drop emotion signal 20% of time so model handles unlabeled inference too.
```

Deep Dive C — DPO + GRPO Two-Stage Alignment

Stage 6 (DPO) cheap-and-fast preference alignment. Stage 7 (GRPO) is the polish. Reward model is WavLM-large features → 3-layer MLP → scalar; trained on 5K seed pairs internally rated by VB + ML-LEAD

before scaling to 100K external.

```
# Stage 6 DPO loss (Rafailov et al., 2023, adapted to speech tokens)
L_DPO = -log sigmoid(
  beta * (log pi(y_w | x) / pi_ref(y_w | x)
    - log pi(y_l | x) / pi_ref(y_l | x))
)
# y_w = preferred audio token sequence, y_l = dispreferred
# beta = 0.1, LR = 1e-7, only backbone trained; decoder frozen

# Stage 7 GRPO (Group Relative Policy Optimization, DeepSeek-style)
# Sample G=8 completions per prompt, normalize rewards within group
advantage_i = (r_i - mean(r_1..r_G)) / std(r_1..r_G)
L_GRPO = - sum_i advantage_i * log pi(y_i | x) + KL(pi || pi_ref)
```

Deep Dive D — Streaming Kernel (the 80ms TTFA recipe)

- bullet Text arrives → PCS-Tok BPE (~1 ms) → fed to Phoneme Transformer incrementally.
- bullet Phoneme TF dynamic look-ahead = 2 phonemes max; flushes phoneme embeddings the moment the first word is complete (~20ms after the first byte of text).
- bullet Slow Backbone runs 6 frames ahead (~480ms of audio buffer) on a separate CUDA stream from text intake — overlaps with phoneme arrival.
- bullet Fast Decoder fires immediately on cb0 token availability; per-frame compute ~6ms on H100.
- bullet Mimi-Lite decoder streams 80ms PCM packets directly to WebSocket — no full-segment buffer.
- bullet KV-cache is reused across user turns in the same conversation — continuation latency drops to ~45ms.
- bullet vLLM scheduler is patched to prioritize the audio token sampling kernel ahead of any pending request batching.

Deep Dive E — Distillation to silkpp-edge (mobile / wearables)

Student model: 1.5B backbone (CSM-1B init) + 8-codebook Mimi-Lite. Distilled via (1) cross-entropy on teacher's cb0 distributions, (2) audio-feature L2 on Mimi acoustic frames, (3) preference-distillation: any pair the teacher prefers, force student to assign higher likelihood. Results: 2.5× speedup, UTMOS ≤ 0.1 worse than teacher. Runs at 0.85× real-time on Pixel 8 Pro, 1.4× real-time on iPhone 15 Pro, ready for offline companion-app deployment.

How we prove we won.

The eval suite, the leaderboard, the receipts.

Evaluation Framework

You cannot beat what you do not measure. The SILK++ Eval Suite is the single source of truth — run nightly against current checkpoint, against last released model, and against all 4 incumbents (where APIs allow). The numbers determine ship / no-ship.

DIMENSION	METRIC	TOOL	TARGET	BEATS
Naturalness	UTMOS v2	sarulab/UTMOS22	≥ 4.25	ElevenLabs v3 (4.05)
Intelligibility	WER vs Whisper-large-v3 ASR	openai/whisper-large-v3	$\leq 5\%$ EN, $\leq 8\%$ HI	Silk (~10% HI)
First-packet latency	TTFA p50/p95	Custom load harness, 1k QPS	≤ 110 ms	ElevenLabs Flash (~135 ms practical IN)
E2E voice-to-voice	User-stop → first AI byte	Real device, Mumbai	≤ 350 ms	Hume EVI 3 (~1.2 s)
Emotion control	Blind 1-5 rating, 30 styles	30-rater panel via Surge	≥ 4.0 avg	Hume EVI 3 (~3.95)
Emotion understanding	Multi-class accuracy, 9 emotions	EmoBank 2.0 eval set	$\geq 80\%$	GPT-4o-realtime (~70%)
Code-switch quality	Blind A/B vs Silk, Hinglish	100 paid Indian raters	$\geq 60\%$ win	Rumik Silk current
Voice clone fidelity	Speaker-cos-sim, 3-sec ref	ECAPA-TDNN cos similarity	≥ 0.85	ElevenLabs IVC (~0.82)
Long-form coherence	5-min consistency rating	Internal human panel	$\geq 4.2 / 5.0$	Silk (~3.9, 2-min limit)
Cost	\$/min, self-hosted, 70% util	Cost dashboard	$\leq \$0.008$	ElevenLabs (\$0.11)

The SILK++ Public Leaderboard (Week 14 launch)

Open eval kit on GitHub: prompts, audio reference set, scoring scripts, results-submission endpoint. Forces competitors to publish or look weak. Apply the same playbook ML community used with LMSys Chatbot Arena — we control the leaderboard, we control the narrative.

Cash, GPUs, and people.

Exact line items. \$520K total. Ship or lose.

Budget — Detailed Line Items

CATEGORY	ITEM	QTY / DURATION	UNIT	TOTAL
Compute	H100 SXM, training (Phase 2-3)	64 × 3 mo reserved	\$2.10/GPU-hr	\$291,000
Compute	H100, codec pretrain burst	8 × 14 days on-demand	\$2.80/GPU-hr	\$7,520
Compute	Inference (preview + GA Wk13-16)	16 H100 × 6 wk	\$2.10/GPU-hr	\$33,800
Compute	Spot fallback budget	buffer	—	\$15,000
Data	Voice actor recordings	50 actors × 20h + studio	\$1,500/actor	\$95,000
Data	DPO/GRPO preference labeling	100K pairs × 5 raters	\$0.18/pair-rater	\$90,000
Data	Synthetic engine compute	OpenAI + ElevenLabs API	—	\$28,000
Hardware	Edge testbed kits	8 mobile dev kits	—	\$25,000
Hardware	Dev workstations	7 × RTX 6000 Ada	\$10K each	\$70,000
Software	W&B Teams, Linear, Lhotse, monitoring	16 wk	—	\$8,000
Software	Eval rater pool (Surge/Karya)	ongoing	—	\$12,000
Legal	NDAs, IP filings, ToS, voice rights	—	—	\$15,000
Marketing	Launch press, demo production, ads	Week 14-16	—	\$25,000
Buffer	Contingency	10%	—	\$52,000
TOTAL				\$520,000+ / 16 wk

Hiring plan (7 seats total, all filled by Wk 5)

SEAT	ROLE	BASE COMP (annualized)	EQUITY	START
1	VB / CTO (you)	—	—	filled
2	ML-LEAD — Principal ML Engineer	■95L + ■40L bonus	0.8%	Wk 1
3	CODEC — Audio Researcher	■70L + ■20L bonus	0.4%	Wk 3
4	DATA — Senior Data Engineer	■60L + ■15L bonus	0.25%	Wk 2
5	INFRA — Senior MLOps Engineer	■65L + ■15L bonus	0.25%	Wk 1
6	RLHF — Alignment Researcher	■85L + ■25L bonus	0.5%	Wk 3

7	FE — Streaming / Mobile Engineer	■55L + ■12L bonus	0.2%	Wk 4
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Sourcing plan: ML-LEAD via direct outreach to ex-Sesame, ex-Fish-Audio, ex-Suno engineers on LinkedIn + GitHub blame log of CSM/Moshi repos. CODEC via DLR audio research community. RLHF via ex-Anthropic / OpenAI alignment alumni — high pay, real research budget, public publication policy. Goal: every offer accepted within 5 days of first call.

What kills us.

And how we kill the risk first.

Risk Register & Mitigations

RISK	SEVERITY	PROBABILITY	MITIGATION
DPO preference data fails inter-rater agreement (kappa < 0.6)	HIGH	30%	Use 5-rater majority; add tie-break rater; Karya pilot 5K pairs in Wk 9 before
Mimi-Lite codec PESQ stuck < 3.9	HIGH	25%	Fall back to vanilla Mimi (no edge mode); ship without 8-codebook mobile f
ML-LEAD hire slips by 4+ weeks	CRITICAL	20%	Backup pipeline of 3 candidates pre-screened; VB acts as interim lead (you
Voice actor studio booking falls through	MED	15%	Booked 2 redundant studios (NeoSound + Sound Ideaz); 4-week buffer; re
ElevenLabs / OpenAI launches a competing Hinglish model in Q3	MED	40%	Move launch 1 week earlier; bank synthetic + actor data ASAP; lock enterp
GPU spot prices spike (CoreWeave / Lambda)	MED	30%	Reserved capacity locked Wk 1; 10% buffer; AWS Trainium fallback validat
Closed-beta users churn / negative TTFA reviews	MED	25%	Aggressive observability; on-call rotation; weekly model push; FE bonus tie
Regulatory: TRAI voice biometrics rules tighten	LOW	15%	Voice clone consent flow built day 1; opt-in only; legal review Wk 8; geo-fer
Architecture choice (CSM-fork) hits licensing snag	LOW	10%	CSM-1B is Apache-2.0; document weight provenance; have clean-room re

The 3 things that will actually kill us (founder's note)

- 1. You hire wrong.** ML-LEAD is the single most important hire. They have to be senior, ego-low, willing to ship in 16 weeks. If they want a 'research environment', they're wrong. Make 8 calls a week until you find the right one. Pay above market.
- 2. You under-spend on data.** Every voice AI startup that lost did so because they spent on GPUs and not on actors. The 1,500-hour studio corpus is the moat. Do not cut it to save \$30K.
- 3. You delay launch.** Voice AI is a winner-take-most category. Whoever lands the Indian consumer companion app first compounds. 03 Sep 2026 is the date. If something slips, cut scope — ship without GRPO, ship without edge mode, ship without Frankfurt region — but ship.

References, repos, commands.

Day-1 setup. Read-list. Source citations.

Appendix

Day-1 / Day-2 commands (run them now)

```
# Day 1
git clone https://github.com/SesameAILabs/csm
git clone https://github.com/fishaudio/fish-speech
git clone https://github.com/kyutai-labs/moshi # contains Mimi reference impl
huggingface-cli download sesame/csm-1b
huggingface-cli download kyutai/mimi
huggingface-cli download fishaudio/fish-speech-1.5

# Day 2 — first end-to-end Hinglish demo
python csm/run.py --text 'main kal milunga, promise!' --voice mum_f02
# baseline UTMOS expected ~3.4 → this is your starting point

# Day 3-7 — fine-tune on 10 hrs Hinglish
python finetune.py --base sesame/csm-1b --data hinglish_seed.tsv --steps 5000
# By end of Wk 1 you have a 'Silk-mini' running on a single 4090.
```

Recommended reading order (5 papers + 2 blogs)

- bullet** Sesame Labs — 'Crossing the Uncanny Valley of Voice' (architecture truth). sesame.com/research/crossing_the_uncanny_valley_of_voice
- bullet** Kyutai — Mimi codec explainer (codec design). kyutai.org/codec-explainer
- bullet** Fish-Audio — Fish-Speech S2-Pro repo (Dual-AR + GRPO recipe). github.com/fishaudio/fish-speech
- bullet** EmoVoice — 'Natural-Language Emotion Control in TTS'. arxiv.org/abs/2504.12867
- bullet** VoXstream — 'Full-Stream TTS with Extremely Low Latency'. arxiv.org/abs/2509.15969
- bullet** Rumik Silk — research blog (your own org's truth doc). rumik.ai/research/silk
- bullet** Hume EVI 3 — launch post (the empathy target). hume.ai/blog/introducing-evi-3

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- [2] Kyutai Mimi codec — kyutai.org/codec-explainer
- [3] Fish-Audio Fish-Speech — github.com/fishaudio/fish-speech
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- [5] VoXstream (Roshchupkina et al., 2025) — arxiv.org/abs/2509.15969
- [6] SpeechAlign — arxiv.org/abs/2404.05600
- [7] Rumik Silk research blog — rumik.ai/research/silk
- [8] Hume EVI 3 launch — hume.ai/blog/introducing-evi-3
- [9] ElevenLabs latency docs — elevenlabs.io/docs/eleven-api/concepts/latency
- [10] Sarvam AI Bulbul — sarvam.ai/apis/text-to-speech
- [11] Hugging Face sesame/csm-1b — huggingface.co/sesame/csm-1b

Ship.

03 September 2026.

No excuses. No extensions. No competitors left standing in your lane.

— Prepared for Vatsal Bharti, CTO, Rumik.ai —

Project SILK++ / Kill-List Battle Plan / v1.0 / Internal Use Only